Energy-Aware Client Selection in Federated Learning as Multi-Round Budgeted Maximum Coverage: Hardness, Exact Models and Battery-Aware Greedy Algorithms

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Abstract—In federated learning (FL) a server repeatedly selects devices that hold data of different groups (e.g., label classes), while the devices differ in speed and battery and every round has a deadline and an energy budget. We formalise multi-round client selection as maximising total weighted group coverage over R rounds subject to deadline, per-round energy and cardinality budgets, and battery dynamics with recharge and overflow. We prove NP-hardness by reduction from Set Cover (already for one round with unit costs), give an exact ILP, and show that single-round greedy with partial enumeration attains the optimal $(1-1/e)$ factor for the knapsack-constrained case. We then show that the myopic round-wise greedy can lose a factor approaching 2 on an adversarial family, and propose a battery-aware greedy with price $e_k(1+\gamma\cdot\text{depletion}_k)$ and a rolling-horizon ILP. On small instances solved exactly, the myopic greedy is 5.2–7.4% below the optimum and the rolling-horizon ILP 0.8–1.8%. On large instances the battery-aware greedy raises coverage by 1.2% (paired Wilcoxon $p<0.001$) and cuts the number of exhausted clients by 37%; coverage correlates with FedAvg accuracy (Pearson $r=0.69$).

***Index Terms—**federated learning, client selection, submodular maximisation, budgeted maximum coverage, integer programming, battery-aware scheduling.*

I. INTRODUCTION

Federated learning (FL) [1] trains a shared model on data that stay on users' devices. In every round the server sends the current model to a subset of clients, collects their updates and aggregates them. *Which* clients take part matters: devices differ in speed and battery, rounds have a deadline, and the data held by the chosen clients decide how representative the update is. Always choosing fast, data-rich devices drains their batteries and biases the model [2]; choosing at random wastes rounds.

We study this as a combinatorial optimisation problem. Each client holds data of a few *groups* (label classes or user segments); the value of a round is the total weight of groups represented by at least one selected client, a monotone submodular coverage function. A round is limited by a deadline, an energy budget and a client cap, and every client owns a battery that participation drains and a known recharge refills. The goal is to maximise coverage over R rounds. Our contributions are:

- **Model and exact ILP** with an exact overflow-slack linearisation of the battery dynamics, an independent solution checker, and a fairness extension (Section III).
- **NP-hardness** by reduction from Set Cover, valid already for one round with unit costs, machine-verified on 632 tiny cases, and the resulting $(1-1/e)$ inapproximability (Section IV).
- **Single-round guarantees**: modified greedy ($\tfrac{1}{2}(1-1/e)$) and partial enumeration $(1-1/e)$ for the knapsack case, with measured ratios on 300 instances (Section V-A, VII-A).
- **Why myopic fails**: an adversarial family on which round-wise greedy achieves exactly $(n+1)/(2n-1) \to \tfrac{1}{2}$ of the optimum, which a battery price cannot repair, but a rolling-horizon ILP can (Section V-B).
- **Battery-aware greedy** with price $e_k(1+\gamma\cdot\text{depletion}_k)$ and its evaluation against the ILP, an upper bound, and baselines, including a FedAvg experiment (Sections V-C, VII).

II. RELATED WORK

FedAvg [1] introduced communication-efficient FL with random client sampling. FedCS [2] observed that heterogeneous resources make random sampling inefficient and selects the largest number of clients that can finish download, update and upload within a deadline. Its objective counts clients; ours maximises weighted *data coverage* and adds battery dynamics across rounds. A single deadline-constrained round is a budgeted maximum coverage problem: Khuller, Moss and Naor [3] give a $(1-1/e)$-approximation by greedy with partial enumeration, building on the analysis of submodular maximisation by Nemhauser, Wolsey and Fisher [4] and on Sviridenko's knapsack-constrained algorithm [5]. These guarantees are per round; with batteries, rounds are coupled and per-round optimality does not compose, which is the gap we study.

III. PROBLEM FORMALISATION

A. Notation and model

Let K be the clients, G the data groups and rounds $t=1,\dots,R$. Client k holds group g iff $h_{kg}=1$; group g has weight w_g. Client k needs time τ_k and energy e_k per participation, has initial battery b^0_k, capacity β_k and recharge $\rho_{kt}\geq 0$ in round t. A round has deadline T_{max}, energy budget E_t and may use at most m clients; $\varrho\in[0,1]$ is a minimum participation fraction ($\varrho=0$ disables fairness). Variables: $x_{kt}\in\{0,1\}$ (selected), z_{gt} (group covered), b_{kt} (battery at the start of round t) and overflow $\omega_{kt}\geq 0$.

$$\max \sum_{t=1}^{R} \sum_{g\in G} w_g\, z_{gt} \tag{1}$$

$$z_{gt} \leq \sum_k h_{kg}\, x_{kt} \quad \forall g,t; \quad x_{kt}=0 \text{ if } \tau_k > T_{max} \tag{2}$$

$$\sum_k x_{kt} \leq m, \quad \sum_k e_k\, x_{kt} \leq E_t \quad \forall t \tag{3}$$

$$e_k\, x_{kt} \leq b_{kt} \quad \forall k,t \tag{4}$$

$$b_{k,t+1} = b_{kt} - e_k x_{kt} + \rho_{kt} - \omega_{kt}, \quad b_{k,t+1} \leq \beta_k \tag{5}$$

$$b_{k1}=b^0_k, \quad \sum_t x_{kt} \geq \lfloor \varrho R \rfloor \text{ (eligible } k) \tag{6}$$

$$x_{kt}\in\{0,1\},\ z_{gt}\in[0,1],\ b_{kt},\ \omega_{kt} \geq 0 \tag{7}$$

Assumptions. (A1) Deadline is a per-client eligibility test, $\tau_k \leq T_{max}$. (A2) e_k is constant; recharge is applied after the round and the battery is capped at β_k. (A3) Recharge is known to the ILP; the greedy policies use only the current battery, hence are also valid when future recharge is unknown. (A4) Coverage is binary: holding any data of a group represents it. (A5)

Holdings h_{kg} and weights are time-invariant. (A6) Fairness (6) applies only to deadline-eligible clients.

Exactness of (5). The original rule $b_{k,t+1}=\min\{\beta_k, b_{kt}-e_k x_{kt}+\rho_{kt}\}$ is equivalent to (5): a larger battery only relaxes (4) and (5) in later rounds, so discarding energy beyond overflow never helps and the optima coincide. Since x is binary, z is integral at an optimum, so z can be continuous, which speeds up the solver. We also tried presolve fixing ($x_{kt}=0$ if $e_k>E_t$) and valid cumulative-energy cuts $\sum_{s\le t} e_k x_{ks} \le b^0{}_k+\sum_{s<t}\rho_{ks}$; they gave no speed-up with HiGHS (Section VII-B).

IV. COMPLEXITY

Theorem 1. The decision problem "is there a selection with total coverage $\ge V$?" is NP-hard in the strong sense, even for $R=1$, unit weights, unit energies, no batteries and no fairness.

Proof. Reduce from Set Cover [6]: universe $U=\{u_1,\dots,u_n\}$, sets $S_1,\dots,S_p$, integer k. Create one group per element ($w=1$) and one client per set with $h_{jg}=1$ iff $u_g\in S_j$; set $e_j=1$, $\tau_j=0$, $T_{max}=1$, $m=E_1=k$, $R=1$, and $\beta_j=b^0{}_j=1$, $\rho=0$. A selection of at most k clients covers all n groups iff k sets cover U; take $V=n$. The construction is polynomial and all numbers are in $\{0,1,k\}$, hence strong NP-hardness. ∎

Verification. We generated random Set Cover instances ($|U|=8$, 8 sets), computed the minimum cover size by brute force and solved the constructed ILP for every k: the two answers agreed on 632/632 (instance, k) pairs.

Corollary 1. With unit costs and $R=1$ the problem is Maximum k-Coverage, so by Feige [8] no polynomial algorithm guarantees $(1-1/e+\varepsilon)$ unless P=NP. Hence the $(1-1/e)$ of Section V-A is optimal. If batteries are unlimited the rounds decouple into R independent budgeted coverage problems; batteries create the coupling that defeats per-round optimality (Section V-B). The pseudo-polynomial knapsack DP does not help because coverage is not additive.

V. ALGORITHMS AND ANALYSIS

A. Single round: greedy with partial enumeration

Fix a round with candidate set A, budget E and cap m. Let $f(S)$ be the weight of groups covered by S (monotone, submodular) and $\Delta(k|S)$ the marginal gain. **ROUND-GREEDY** (Algorithm 1) repeatedly adds the affordable client with the largest $\Delta(k|S)/c_k$, where c_k is a *price* (the true energy e_k by default); feasibility always uses e_k. The returned set is the better of the greedy set and the best single affordable client. **PARTIAL-ENUM(3)** tries every start set of at most three clients and completes each with ROUND-GREEDY.

```
Alg. 1  ROUND-GREEDY(A, c, E, m)
1  S <- {}
2  while |S| < m:
3    C <- {k in A\S : e_k <= E - e(S)}
4    k* <- argmax_{k in C} gain(k|S) / c_k
5    if gain(k*|S) = 0: break
6    S <- S + {k*}
7  return better of S and best single client
```

Theorem 2. For the knapsack constraint only ($m=\infty$) and prices $c=e$: (a) ROUND-GREEDY with the best-single fallback is a $\tfrac12(1-1/e)$-approximation; (b) PARTIAL-ENUM(3) is a $(1-1/e)$-approximation [3], [5]. For the cardinality constraint only ($c\equiv1$, $E=\infty$) plain greedy is a $(1-1/e)$-approximation [4].

Proof of (a). Let O be an optimum with $e(O)\le E$ and let G_i be the first i clients picked by the idealised greedy that ignores the budget, with costs c_i. Let $l+1$ be the first pick that does not fit, so G_l fits and $\sum_{i\le l+1} c_i>E$. By submodularity, $f(O)-f(G_{i-1}) \le \sum_{o\in O\setminus G_{i-1}} \Delta(o|G_{i-1}) \le (\Delta_i/c_i)\cdot E$, hence $f(G_i)-f(G_{i-1}) \ge (c_i/E)(f(O)-f(G_{i-1}))$ and $f(G_{l+1}) \ge [1-\prod_{i\le l+1}(1-c_i/E)]\, f(O) > (1-1/e)\, f(O)$, using $\sum c_i>E$ and the AM–GM inequality. Because $f(G_{l+1}) \le f(G_l)+f(\{g_{l+1}\})$, one of G_l and the single client g_{l+1} (which fits alone) has value $\ge \tfrac12(1-1/e)f(O)$. Our implementation skips non-fitting clients and continues; by monotonicity its set is a superset of G_l, so the bound holds. For (b), guessing the three highest-gain clients of O makes the first non-fitting pick contribute at most $f(O)/3$, which removes the factor ½ [3], [5]; (c) is the same argument with $c_i/E=1/m$ [4]. ∎

The proof does not cover *both* a knapsack *and* a cardinality constraint, which is our actual setting; we therefore measure ratios empirically (Section VII-A).

B. Why the myopic greedy fails

Proposition 3. For every n there is a two-round instance on which the round-wise greedy earns exactly $(n+1)/(2n-1)$ of the optimum, which tends to ½.

Proof. Unit weights, $m=1$, $E=(2,1)$. Client A covers all n groups, $e=1$, battery for exactly one participation, $\rho=0$. Client B covers $n-1$ groups, $e=2$, battery for one participation. Client W covers one group, $e=1$, unlimited battery. In round 1 greedy takes A (best gain and best ratio n vs $(n-1)/2$). In round 2 B is unaffordable (2>1) and A is exhausted, so W is taken: total $n+1$. The optimum takes B in round 1 and A in round 2: $2n-1$. ∎

A and B have identical depletion before round 1, so any price that depends on the battery level alone cannot separate them; fixing this requires knowing the future budget E_2.

C. Battery-aware greedy

BATTERY-AWARE (Algorithm 2) runs ROUND-GREEDY in every round with price $c_k=e_k(1+\gamma\cdot\text{depletion}_k)$ and $\text{depletion}_k=1-b_{kt}/\beta_k\in[0,1]$, where $\gamma\ge0$ is tuned. A drained client is "expensive", so the greedy spreads participation, but the budget test still uses the true e_k. $\gamma=0$ recovers the myopic greedy.

```
Alg. 2  BATTERY-AWARE(gamma)
1  b <- b0
2  for t = 1..R:
3    A <- {k : tau_k <= Tmax and b_k >= e_k}
4    c_k <- e_k  (1 + gamma  (1 - b_k/beta_k))
5    S_t <- ROUND-GREEDY(A, c, E_t, m)
6    b_k <- min(beta_k, b_k - e_k*[k in S_t] +
rho_kt)
```

Lemma 4. (i) Every schedule produced is feasible for (2)–(5). (ii) The running time is $O(R\cdot m\cdot|K|\cdot|G|)$. (iii) If $b_{kt}=\beta_k$ and $\rho_{kt}\ge e_k$, then participating leaves the next battery at β_k, i.e. it is free, and the price is the lowest possible (depletion 0).

Proof. (i) The candidate set enforces deadline and battery, ROUND-GREEDY enforces E_t and m, and batteries follow (5). (ii) Each of at most m iterations scores $\le|K|$ clients over $|G|$ groups. (iii) $\min\{\beta,\beta-e+\rho\}=\beta$ iff $\rho\ge e$. ∎

Lemma 4(iii) explains the price: clients near full capacity would waste recharge as overflow anyway, so using them is cheap; drained clients carry a real opportunity cost.

D. Rolling-horizon ILP and an upper bound

ROLLING-ILP(W) solves the ILP (1)–(7) for rounds $t,\dots,t+W-1$ from the current batteries (recharge in the window assumed known) and commits only round t; $W=1$ is the per-

round optimum and $W=R$ is exact. **Lemma 5.** Let $UB=\sum_t OPT_t$, where OPT_t is the optimum of round t ignoring batteries. Then $UB \geq OPT$. *Proof:* dropping (4)–(5) relaxes the problem and decouples rounds. ∎ We use UB when the full ILP is too large.

VI. EXPERIMENTAL SETUP

Instances. Each client holds 2–4 random groups of G; weights are integers in 1..5; τ_k is log-normal(0, 0.5); $e_k=10\tau_k/\text{mean}(\tau)$; T_{max} is the 70% quantile of τ; $\beta_k=3e_k$, $b^0_k \sim U(0.5,1)\beta_k$; recharge is $r_k \cdot e_k \cdot U(0.5,1.5)$ with $r_k \sim U(0.05,0.30)$, so a client can sustain about r_k participations per round; $E_t=m\cdot\text{mean}(e)\cdot U(0.8,1.2)$ and $m=\max(3,\lfloor G/3 \rfloor)$ unless stated. **Methods.** Random, Fastest-first, Myopic greedy (γ=0), Battery-aware (γ=1.5), Rolling ILP (W=3). **Metrics.** Coverage (also as % of $R\cdot\sum w$), fraction of rounds with every group covered, *exhausted clients* (eligible clients that at least once could not afford e_k), and Jain's index [9] of participation counts of eligible clients. **Tooling.** Python 3, NumPy/SciPy [11], ILPs solved by HiGHS [10] through `scipy.optimize.milp`, one CPU core, fixed seeds. Every schedule is re-verified by an independent checker that recomputes batteries and objective from the 0/1 matrix.

VII. RESULTS

A. ILP validation and single-round ratios

The single-round ILP matched brute-force enumeration on all 60 tiny instances (0 mismatches). Table I gives the ratio to the ILP optimum on 150 random instances per setting (K=26, G=15, budget = 25% of eligible energy). PARTIAL-ENUM(3) was optimal on every instance; ROUND-GREEDY with fallback never fell below 0.85, far above the worst-case $\frac{1}{2}(1-1/e)\approx 0.316$; plain cardinality greedy fell to 0.77. Running times were 0.11 ms (modified greedy) and 34.2 ms (enumeration). The LP relaxation was on average 1.9% above the optimum (max 6.9%). Ratio-greedy alone is not enough: on the instance {cost 1, value 1; cost L, value $L-1$}, budget L, it earns 1 while the optimum is $L-1$, which the best-single fallback repairs.

TABLE I. Single-round value / ILP optimum (150 instances per setting).

Algorithm	Knapsack mean	min	+cap. m=5 mean	min
Plain greedy	0.946	0.773	0.948	0.773
Ratio greedy	0.975	0.849	0.958	0.849
Modified greedy	0.975	0.849	0.958	0.849
Partial enum. (3)	1.000	1.000	1.000	1.000

B. Small multi-round instances against the exact ILP

Table II compares policies with the exact multi-round ILP on 8 seeds per size (30 s limit; every instance with $K \leq 20$ was solved to optimality, 5/8 for K=30 with a proven bound gap of at most 2.1%, so reported gaps are relative to the incumbent). Random and Fastest lose 18–27%. Myopic greedy loses 5.2–7.4%; the battery-aware variant is not significantly different from it on these tiny instances (Wilcoxon p=0.20). Rolling ILP is within 1.8% on average. The LP relaxation is 3.2–5.1% above the optimum. Presolve and cumulative cuts did not help (mean solve time 18.2 s with vs 16.6 s without for K=30).

TABLE II. Mean gap to the exact ILP (8 seeds per size).

Policy (gap to ILP, %)	K=15,R=5	K=20,R=8	K=30,R=10
Random	20.9	18.4	27.0
Fastest	21.7	20.3	26.5
Myopic greedy	5.2	7.4	7.2
Battery-aware	6.1	6.0	6.2
Rolling ILP (W=3)	0.9	1.8	0.8

C. Large instances

Table III uses 15 seeds of K=100, G=30, R=30 (A) and K=60, G=12, R=20 (B), measured against the battery-free upper bound UB (Rolling ILP, 3 s per window, only 5 seeds; on the same 5 seeds in A it reaches 93.5% against 88.8% for battery-aware and 87.8% for myopic). In A the battery-aware greedy improves on myopic by 1.2% on average, winning 14/15 instances (Wilcoxon p<0.001), and reduces exhausted clients from 40.2 to 25.3; in B the coverage gain is smaller (0.3%, p=0.08) but full-coverage rounds rise from 0.74 to 0.80 and exhausted clients halve. The heuristics need about 13 ms for K=100 and 47 ms for K=800 over 30 rounds, versus 34 s for Rolling ILP.

TABLE III. Large instances (means; *Rolling ILP: 5 seeds).

Policy	%UB (A)	Exh. (A)	Jain (A)	%UB (B)	Full (B)	Exh. (B)
Random	63.0	36.7	0.87	74.6	0.01	11.7
Fastest	64.7	47.8	0.63	75.0	0.02	20.4
Myopic	89.1	40.2	0.66	98.5	0.74	16.4
Batt.-aware	90.2	25.3	0.69	98.8	0.80	7.7
Rolling ILP*	93.5	38.6	0.69	100.0	1.00	4.0

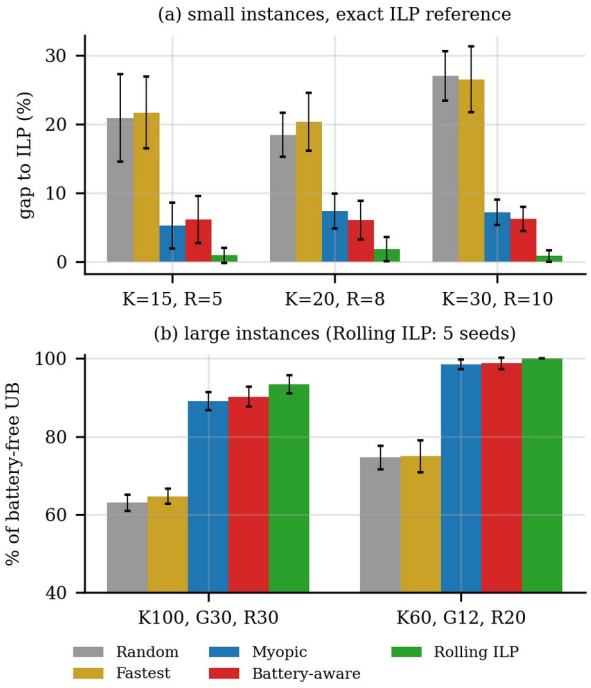


Fig. 1. (a) Gap to the exact ILP on small instances; (b) coverage relative to the upper bound on large instances (error bars: one standard deviation).

D. Tuning γ

Fig. 2 sweeps γ over 30 seeds. In the large-G setting coverage rises from 75.1% (γ=0) to 77.1% at γ=1.5 (p<0.001) and plateaus; exhausted clients fall monotonically from 36.0 to 24.0 and Jain's index rises. In the small-G setting coverage peaks near γ≈0.5–1 (95.4% vs 94.9%) and then declines to 92.7% at γ=8, while exhausted clients fall from 14.2 to 4.2 and Jain's index from 0.65 up to 0.79. Thus γ trades a little coverage for fairness and battery health; γ∈[1,2] is a good default.

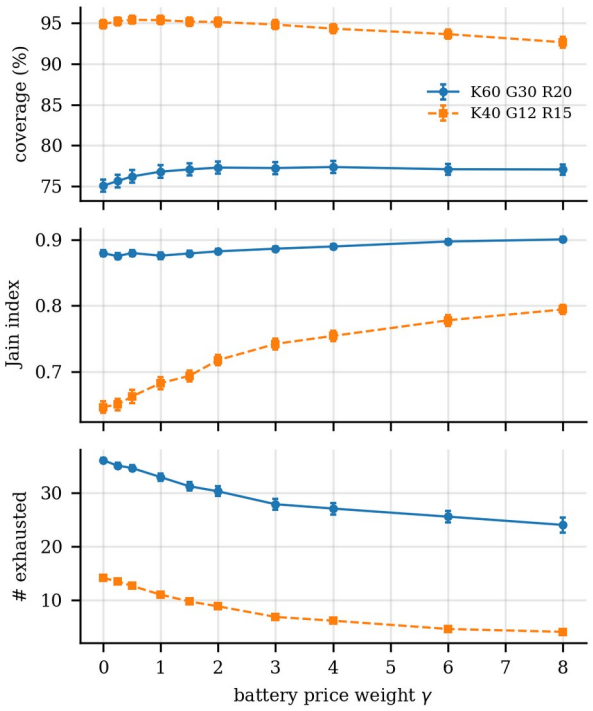


Fig. 2. Effect of the battery-price weight γ (30 seeds, mean ± s.e.).

E. Adversarial family and energy scarcity

On the family of Proposition 3 the measured ratios of the myopic greedy are 0.667, 0.579, 0.538, 0.515, 0.508 and 0.501 for n=5,…,1000, exactly $(n+1)/(2n-1)$. The battery-aware greedy produces the identical schedule (ratio 0.501), as predicted, whereas Rolling ILP with W=2 is optimal (Fig. 3a). Fig. 3b varies recharge from 0.25× to 4×. The battery-aware gain over myopic is largest under moderate scarcity (2.0 points at 1×, with exhausted clients 19.8 → 10.8 at 2×) and vanishes when energy is very scarce (every client is drained) or abundant (batteries never bind), which delimits when battery awareness matters.

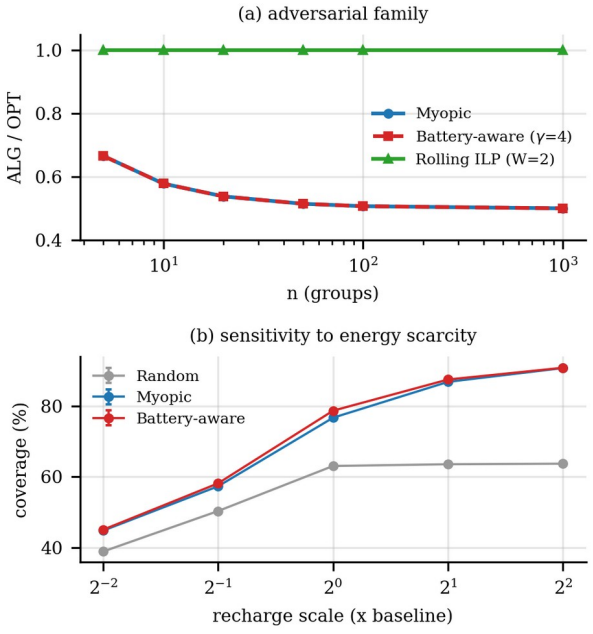


Fig. 3. (a) ALG/OPT on the adversarial family; (b) coverage vs recharge scale.

F. Fairness variant

With the minimum participation constraint (6) on K=20, R=10 (8 seeds, ILP), forcing $\varrho \in \{0, 0.1, 0.2\}$ costs no coverage, ϱ=0.3 costs 0.13%, ϱ=0.4 is feasible for only 2/8 instances (loss 5.1% among them), and ϱ=0.5 is infeasible for all, because batteries cannot sustain that participation rate.

G. Link to FL accuracy

We ran FedAvg (softmax regression, 2 local epochs) on the scikit-learn 8×8 digits data [12] as a stand-in for MNIST (no internet access); the 10 classes are the groups and each client holds exactly its groups' classes. Over 192 runs (4 policies × 4 values of m × 12 seeds) mean per-round coverage and final test accuracy are strongly correlated (Pearson r=0.69, Spearman ρ=0.74; Fig. 4); accuracy rises from 80.3% at 57.1% coverage (m=2) to 89.0% at 92.6% (m=6). Coverage-driven policies (Myopic 87.5%, Battery-aware 87.8%) beat Random (84.3%) and Fastest (82.2%); the Myopic vs Battery-aware accuracy difference is within noise.

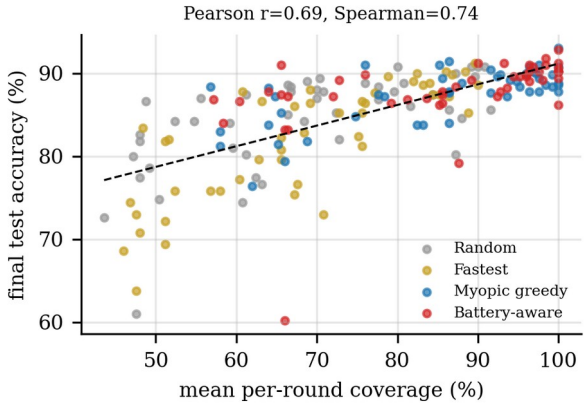


Fig. 4. Per-round coverage vs final FedAvg accuracy (dashed: linear fit).

VIII. DISCUSSION AND LIMITATIONS

The battery price is a cheap, online-in-ρ improvement: it needs no forecast of recharge and about 10 ms per run, and it consistently reduces exhausted clients, but its coverage gain is modest (about 1–2%) and it cannot detect budget-driven traps (Proposition 3). Rolling-horizon ILP is near-optimal but needs known recharge/budgets and seconds per decision. Limitations: (i) Theorem 2 does not cover the combined knapsack+cardinality setting; (ii) the exact ILP is solved only up to about 30 clients and 10 rounds within 30 s; (iii) instances are synthetic, and the FL experiment uses a small linear model on the digits data rather than MNIST; (iv) coverage is binary (the probabilistic-coverage and online-recharge variants are left as future work, although the greedy policies are already online in ρ).

IX. CONCLUSION

Multi-round energy-aware client selection is a strongly NP-hard coverage problem whose per-round core admits an optimal-factor greedy, but whose battery coupling defeats myopic selection by up to a factor approaching two. A battery-aware price improves coverage and markedly reduces exhausted clients at negligible cost, and a rolling-horizon ILP comes within about 1–2% of the exact optimum. Future work includes guarantees for the two-constraint round problem, probabilistic coverage, and unknown-recharge competitive analysis.

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